Vijay Mohanram Iyer

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iyervijay99@gmail.com +49-17667345305 Vijay's_Portfolio Vijay Iyer

SUMMARY

Graduate engineer specializing in intelligent safety systems and health applications powered by computer vision and machine learning. Skilled in transforming advanced algorithms including deep learning architectures and large language models into practical, user-centered tools that enhance security and promote human well-being.

PROFESSIONAL EXPERIENCE

1. FZI — *September 2025 – Present*
Research Assistant

- Evaluate and optmize detection models through transfer learning and ensemble methods for hyperparameter fine-tuning and multimodal data analysis.
- Optimize detection models to enhance performance on challenging visual manipulations and research publication.
- Conduct literature review and work towards research publication.

2. TecoLab — *March 2023 – Present*
Working Student

- Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.
- ML4Print: Classification based on printer-specific characteristics and paper substrate type using feature extraction and supervised ML classifiers.
- Computational Fluid Dynamics: utilized regression models and data integration to simulate the thermal behavior of liquids within industrial valves, thereby enhancing predictive maintenance capabilities and system reliability.

3. FZI — *February 2025 – August 2025*
Master Thesis

- Developed an end-to-end deep learning pipeline using rPPG signals for artifact evaluation and Benchmarked under real-world conditions.
- Proposed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification and acquired Benchmark results under real-world conditions.

4. Access@KIT — *March 2023 – May 2023*
Working Student

- Developed a bi-directional LSTM-based text-to-speech system with phoneme alignment and acoustic feature extraction, optimizing temporal dependencies and prosody for accessible, natural speech for visually impaired users.

5. Accenture India Pvt. Ltd. *February 2022 – April 2022*
Associate Software Engineer

- Maintained IBM Data-center servers with batch job scheduling and performance monitoring COBOL scripts.

EDUCATION

M.Sc. in Electrical Engineering and Information Technology *May 2022 – July 2025*
Karlsruhe Institute of Technology
GPA: 2.3

Bachelor of Engineering in Electronics Engineering *August 2017 – May 2021*
University of Mumbai, India
GPA: 2.8

PROJECTS

1. **Attention Mechanism and Multi-ROI Artifact Recognition for DeepFake Detection Using rPPG (FZI - Thesis)** *February 2025 – July 2025*
   Utilized POS and CWT to extract pulsatile component for the identification of irregularity in signals for real and forged videos and enhanced privacy and security.
   Technology: Python, PyTorch, OpenCV, NumPy, Matplotlib, CNN, Vision Transformer, rPPG
2. **CamCussion: Eye Tracking Software (Zeiss Innovation Hub)** *November 2023 – July 2024*
   Utilized computer-vision to analyze pupil dilation and saccadic eye movements in real time to track and assess eye behavior, contributing to accurate concussion diagnosis.
   Technology: Python, OpenCV, Dlib, PyQt, Real-time image processing
3. **Self-Driving Car using LIDAR** *September 2022 – February 2023*
   Developed a real-time 3D obstacle detection using LIDAR data in autonomous driving. It transforms raw point clouds into ViT-compatible tokens via efficient voxelization and applies Multi-head attention to capture global context, improving detection accuracy.
   Technology: Python, PyTorch, Open3D, NumPy, Vision Transformer, LiDAR processing
4. **Real-Time Car Accident Alert System** *January 2022 – june 2022*
   Developed an embedded LSTM-based accident detection system running on ESP32 using TensorFlow Lite for Microcontrollers, enabling real-time crash detection and automated SMS/email alerts with precise GPS location.
   Technology: Python, TensorFlow Lite, ESP32, LSTM, Embedded Systems

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, ROS2, React.js
- **ADAS:** Sensor fusion, camera and LiDAR calibration, perception algorithms, path planning, object detection & tracking
- **Tools:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX
- **Spoken Languages:** English C1, German A2 (Learning)

SOFT SKILLS

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging, Balancing and Prioritizing tasks to meet goals.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.